

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING

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COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Application-Based Questions

Weightage: 40% QUESTIONS: Total Marks: 30

1: Online Symposium Portal

A college is developing an online symposium registration portal. The following HTML structure and HTTP interaction is observed:

```
<form action="/register" method="POST">
  <input type="text" name="name">
  <input type="email" name="email">
  <button>Submit</button>
</form>
```

When a student submits the form, the browser sends an HTTP request to the server, and the server responds accordingly.

Questions:

1. Identify appropriate **HTML5 semantic elements** missing in the structure.
2. Modify the structure to make it **standards-compliant**.
3. Interpret the **HTTP request generated** during form submission.
4. Analyze the **HTTP response cycle** from server to browser.
5. Suggest improvements for **usability and accessibility**.

2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet is used:

```
<html>
<head>
  <title>Company</title>
</head>
<body>
  <div>
    <h1>Welcome
    <p>About us
  </div>
</body>
```

</html>

Questions:

1. Identify **improper nesting and missing closing tags**.
2. Analyze how different browsers may interpret this structure.
3. Identify **missing semantic elements** (e.g., header, section).
4. Provide a **corrected W3C-compliant HTML structure**.
5. Suggest techniques to ensure **cross-browser compatibility**.

3: Form Submission Failure

A web application fails to send form data to the server. The following configuration is observed:

```
<form action="/submit" method="GET">  
  <input type="text" name="username">  
  <input type="password" name="password">  
  <button type="button">Login</button>  
</form>
```

Questions:

1. Identify issues in the **form configuration**.
2. Analyze why the **HTTP request is not triggered**.
3. Interpret the role of **GET method in this scenario**.
4. Propose a **corrected implementation**.
5. Suggest improvements for **secure data transmission**.

Code of Conduct:

I Lakshmi(192373066) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: LAKSHMI ANKALA
RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

1. Online Symposium Portal

Given HTML Code

```
<form action="/register" method="POST">
  <input type="text" name="name">
  <input type="email" name="email">
  <button>Submit</button>
</form>
```

➤ **Identify appropriate HTML5 semantic elements missing in the structure.**

The following HTML5 semantic elements are missing:

- <header> – Displays the page heading.
- <main> – Contains the main content.
- <section> – Groups related content.
- <form> should include <label> elements for each input field.
- <footer> – Displays footer information.

➤ **Modify the structure to make it standards-compliant.**

```
<!DOCTYPE html>
<html>
<head>
  <title>Symposium Registration</title>
</head>
<body>
<header>
  <h1>Online Symposium Registration</h1>
</header>
<main>
<section>
  <form action="/register" method="POST">
    <label for="name">Name:</label>
    <input type="text" id="name" name="name" required>
    <br><br>
    <label for="email">Email:</label>
    <input type="email" id="email" name="email" required><br><br>
    <button type="submit">Submit</button>
  </form>
</section>
</main>
<footer>
```

<p>© 2026 SIMATS Engineering</p>
</footer>
</body>
</html>

➤ **Interpret the HTTP request generated during form submission.**

When the user clicks **Submit**:

- The browser collects the form data.
- It sends an **HTTP POST request** to:

/register

Example request:

POST /register HTTP/1.1

Host: example.com

Content-Type: application/x-www-form-urlencoded

name=John&email=john@example.com

The data is sent inside the **request body**.

➤ **Analyze the HTTP response cycle from server to browser.**

1. User fills the form.
2. Browser sends an HTTP POST request.
3. Server receives the request.
4. Server validates the submitted data.
5. Server stores the registration details in the database.
6. Server sends an HTTP response.
7. Browser receives the response and displays either:
 - Registration Successful
 - Error message (if validation fails)

➤ **5. Suggest improvements for usability and accessibility.**

- Use <label> for every input field.
- Add the required attribute.
- Display validation messages.
- Use meaningful placeholders.
- Use HTTPS for secure communication.
- Add proper headings and semantic HTML.
- Ensure keyboard accessibility.

2. Cross-Browser Compatibility Issue

Given HTML Code

```
<html>
<head>
<title>Company</title>
</head>
<body>
<div>
<h1>Welcome
<p>About us
</div>
</body>
</html>
```

➤ **Identify improper nesting and missing closing tags.**

- Missing closing tags:
- `</h1>`
- `</p>`

Improper structure:

- HTML elements are left unclosed.
- Missing HTML5 semantic elements.

➤ **Analyze how different browsers may interpret this structure.**

Modern browsers automatically insert missing closing tags.

Example interpretation:

```
<h1>Welcome</h1>
```

```
<p>About us</p>
```

Although the page may display correctly:

- HTML is invalid.
- Different browsers may render spacing differently.
- CSS layouts may behave unexpectedly.

➤ **Identify missing semantic elements.**

The following semantic elements are missing:

- <header>
- <main>
- <section>
- <footer>

➤ **Provide a corrected W3C-compliant HTML structure.**

```
<!DOCTYPE html>
<html>
<head>
<title>Company</title>
</head>
<body>
<header>
<h1>Welcome</h1>
</header>
<main>
<section>
<p>About us</p>
</section>
</main>
<footer>
<p>Copyright 2026</p>
</footer>
</body>
</html>
```

➤ **Suggest techniques to ensure cross-browser compatibility.**

- Use valid HTML5.
- Validate code using the W3C Validator.
- Use CSS Reset or Normalize.css.
- Test on Chrome, Firefox, Edge, and Safari.
- Use responsive design.
- Avoid deprecated HTML tags.
- Use standard CSS properties.

3. Form Submission Failure

Given HTML Code

```
<form action="/submit" method="GET">  
<input type="text" name="username">  
<input type="password" name="password">  
<button type="button">Login</button>  
</form>
```

➤ Identify issues in the form configuration.

Issues:

- The button type is **button** instead of **submit**.
- GET method is not suitable for login.
- No labels are provided.
- Password is exposed in the URL when using GET.

➤ Analyze why the HTTP request is not triggered.

The button is:

```
<button type="button">
```

This button does **not** submit the form.

Only

```
<button type="submit">
```

or

```
<input type="submit">
```

will send the HTTP request.

➤ Interpret the role of GET method in this scenario.

GET sends form data through the URL.

Example:

```
/submit?username=John&password=1234
```

Problems:

- Password becomes visible.
- Browser history stores the URL.
- Server logs may record the password.
- Not suitable for login forms.

4. Propose a corrected implementation.

```
<form action="/submit" method="POST">
<label for="username">Username</label>
<input
type="text"
id="username"
name="username"
required>
<br><br>
<label for="password">Password</label>
<input
type="password"
id="password"
name="password"
required>
<br><br>
<button type="submit">
Login
</button>
</form>
```

5. Suggest improvements for secure data transmission.

- Use **POST** instead of GET.
- Enable **HTTPS**.
- Encrypt passwords before storing them.
- Validate all user inputs.
- Use strong password policies.
- Implement session management.
- Enable Multi-Factor Authentication (MFA).
- Protect against SQL Injection and Cross-Site Scripting (XSS).

SUMMARY TABLE

Question	Key Answer
1. Online Symposium Portal	Use semantic HTML (<header>, <main>, <section>, <footer>), labels, and POST for form submission.
2. Cross-Browser Compatibility	Close all HTML tags, use semantic HTML5 elements, validate with W3C, and test across browsers.
3. Form Submission Failure	Change button type to submit, use POST, and enable HTTPS for secure login.